

Lab 1: PCA, GLM-PCA, and NMF

The goal of this lab (and future labs) is to reflect the process of doing research. Therefore, the questions will be open-ended. There is no one right answer; instead, explain and justify the decisions you make as you do your analysis.

Important notes:

You can (and should) use a coding agent to assist you during your lab. Keep in mind that these agents are prone to shortcuts and hallucinations, so you must understand what is actually happening during your analysis. Throughout the lab, make notes on how you used a coding agent. Try to independently verify interpretations made by the agent by doing literature review. Another (sometimes, more robust) option is to start with validated findings from literature and explore them using the agent.

Final Submission

Submit a reproducible notebook. (Notebooks allow you to divide up code blocks with your own written findings/analysis and plot figures throughout. They are also easier for you to run, and for us to grade).

Within your notebook, answer all the following questions in order.

When relevant, include links and citations for both datasets and their associated papers, with a brief description of each dataset.

Part 1: Single cell RNA sequencing

Part 1.1: Find a dataset

Find a publicly available scRNA seq dataset associated with a published paper/preprint.

Here are some possible data sources:

- Human Cell Atlas
- 10x Genomics public datasets
- NCBI Gene Expression Omnibus
- Broad Institute Single Cell Portal

Your dataset should include:

- A raw (positive) count matrix
- At least 500 cells and 1000 genes
 - Not be *too* large (you should be able to analyze it with your computer or an accessible GPU, like the Google Colab T4)
- Cell type labels for each cell

Briefly describe:

- where you found the dataset
- the biological question studied in the associated paper
- the tissue/biological system studied
- the assay/technology used to collect the data
- the dataset dimensions
- any available annotations

Ideally, you should analyze data from a relatively new dataset/technology (>2021)

Part 1.2: Explore/pre-process the data

Load the count matrix into your environment and follow the standard single cell quality control filtering steps (see Scanpy tutorial here <https://scanpy.readthedocs.io/en/stable/tutorials/basics/clustering.html>). How does the data size, sparsity, etc. change from filtering? (It may help to visualize a histogram of the number counts per gene or cell)

1.3 PCA

Implement the standard PCA pre-processing on your dataset following <https://scanpy.readthedocs.io/en/stable/tutorials/basics/clustering.html>

Visualize the first few principal components. Think about how to visualize these effectively in the context of the dataset you are using.

From your visualization, answer whether the PCs correspond with different cell types? Or do any PCs correspond with any technical artifacts (e.g. fraction of zero counts, as in the GLMPCA paper; library size, i.e. total count for a given cell; etc)?

1.4 GLM-PCA

Run GLM-PCA on the raw count matrix (again, check for existing methods and cite your sources!). Choose a count distribution that you believe is appropriate for the dataset and explain your choice.

Visualize at least the first three GLM-PCA components in a way that is effective in the context of your dataset. One idea is to make these visualizations in comparison to the PCs from 1.3.

Answer the following:

- Do the GLM-PCA components also correspond with different cell types?
 - Or do any components correspond with technical artifacts?
- Which method (PCA/GLM-PCA) performs “better?” (How will you choose to define “better?”)

1.5 NMF

Apply nonnegative matrix factorization (NMF) to the single-cell dataset. You may want to either fix the number K of factors, or “sweep” through several choices of the number K of factors and choose the “best” one (what is “best”? how is the number of factors typically chosen?).

Visualize these factor scores across cells and genes that have the largest factor loadings. Do the factors or loadings correspond to cell types?

Compare NMF with PCA and GLM-PCA. Consider:

- Which method is easiest to interpret?
- Which method is slowest/fastest?
- Are any of these methods good at visualizing a continuous variation of some value between cells?
- Are any of these methods affected by technical variation? (i.e. a PC/factor is correlated to technical variation rather than biological variation)

Part 2: Spatial transcriptomics (ST)

Parts 2.1 / 2.2: Find/QC a dataset

Find a publicly available ST seq dataset associated with a published paper/preprint and perform standard QC. See Parts 1.1/1.2 for more details.

Ideally, you should analyze data from a relatively new dataset/technology (>2023).

2.3 PCA, GLM-PCA, and NMF on the ST dataset

Run both PCA, GLM-PCA, and NMF on the ST dataset.

Visualize the different PCs, GLM-PCs, or factors in the physical 2-D space. Interpret each component you visualize. Consider:

- Do different components correspond to known tissue regions?
- Does any component show a spatial gradient or reflect sparsity?
- How would you quantify any of the spatial patterns found?

Compare the different dimensionality reduction methods. Which method is “better”? (What does “better” mean?). Consider:

- Which method produces more recognizable tissue structures?
- Does any method produce more “noisy” or “smooth” components?

Part 3: Final reflection

Answer this how you see fit.

Some possible directions include:

- What did these different methods reveal in the single cell and ST datasets?
- Was any method substantially affected by sparsity or other technical factors? Why do you think not/so?
- Under what circumstances would you choose PCA, GLM-PCA, or NMF in future analysis?
- Were there any cool/weird findings? Anything interesting/thought provoking?